Post-Double Selection Lasso

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Spring 2026

Motivation

- What if we want to
  - predict treatment effects on observational data?
  - improve precision of RCT treatment effects?
  - or correct for randomization failures?

- You include controls.

- But what if you have dozens or even hundreds of controls?
- What is a scientific procedure to choose controls objectively?

- We can use Post-Double Selection (PDS) Lasso!

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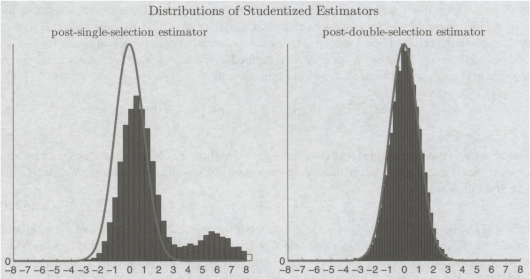
Original Purpose: Observational Data

- Born from Big Data
- Observational datasets have endogenous treatment.
- Leverage rich datasets to orthogonalize treatment.
- Laid the foundation for modern techniques such as Double Machine Learning.



(Belloni et al., 2014)

Modern Application: RCTs

PDS Lasso is primarily used to ameliorate two common issues in RCTs:

1. Randomization failure.
2. Attrition induced bias.

In practice it is useful when:

1. Small samples increase risk of imbalance.
2. There is differential attrition.
3. Outcomes are highly auto-regressive but we do not have lagged outcomes.
4. You have a large set of possible covariates.
   - You want to reduce the number of covariates, but don't strongly care which covariates appear in the model.

What are Linear Regularized Models?

Starting from OLS

- Regression with extra steps.
- Add a penalty term to balance model fit with model complexity.

Core Assumptions & Goals

1. The true model is *sparse*
   - Only a subset of covariates significantly impact on outcomes.
2. We want a simpler model for interpretation.

What is Lasso?

$$\min_{\beta} \left\{ \underbrace{\sum_{i=1}^{n} (Y_i - X_i\beta)^2}_{\text{model fit}} + \lambda \underbrace{\sum_{k=1}^{p} |\beta_k|}_{\substack{\text{model} \\ \text{complexity}}} \right\}$$

$\lambda \equiv$ tuning parameter

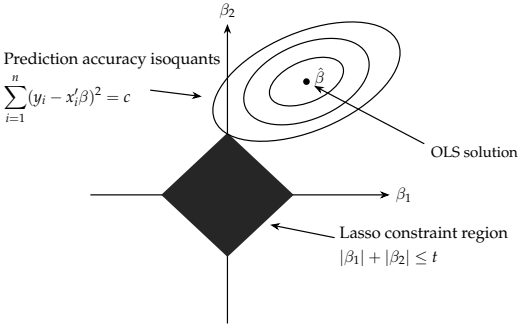
$$\boxed{\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta}, \theta)^2}$$

- **Bias ↑:** shrinkage toward zero, some true effects removed
- **Variance ↓:** fewer covariates, more stable estimates
- **Tradeoff:** variance reduction > bias increase ⇒ lower MSE
- **Interpretation:** sparsity improves prediction, not structural identification

What Does Lasso Look Like?

Choose λ to minimize out-of-sample MSE
i.e. *cross-validation*

$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i'\beta)^2 \quad \text{s.t.} \quad |\beta_1| + |\beta_2| \leq t$$



- **Large λ**
  - Strong penalty
  - Small diamond (tight constraint)
  - More shrinkage, more zeros
- **Small λ**
  - Weak penalty
  - Large diamond
  - Closer to OLS

Benefits of Lasso

1. Selects highly predictive covariates — improves out-of-sample performance
2. Reduces number of covariates in model — more interpretable and stable
3. *In theory*, Lasso has the Oracle Property
   - It perfectly selects the true *sparse* covariates and efficiently converges to the true parameter β.
   - In practice, Lasso fails due to needing strong assumptions but more robust specifications exist, such as Adaptive Lasso & Generalized Outcome-Adaptive Lasso.

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What is Post-Double Selection (PDS) Lasso?

It's very simple!

Suppose we have outcome Y_i, treatment τ_i, and a large control vector X_i.

1. Estimate Lasso $\tau_i \sim X_i$, and store control vector X_τ.
2. Estimate Lasso $Y_i \sim X_i$, and store control vector X_Y.
3. Rerun OLS as $Y_i \sim \tau_i\theta + (X_\tau \cup X_Y)\beta$.
4. That's it, interpret the treatment effect as normal!

What does Double Selection Do?

$$Y_i \sim \tau_i \theta + (X_\tau \cup X_Y)\beta$$

$X_\tau \setminus X_Y \equiv$ Only predictive of treatment

Selection Bias

$X_Y \setminus X_\tau \equiv$ Only predictive of outcome

Treatment Effect Precision

$X_Y \cap X_\tau \equiv$ Predictive of outcomes and treatment

Endogeneity Bias

In principle this increases model power!

Best Practices

1. Always **standardize** variable scale prior to Lasso.
   1.1 The penalty parameter is biased against small covariates.
   1.2 e.g. $n(x) := \dfrac{n - \bar{n}}{SD(n)}$

2. **Missing Observations:** Include controls with missing values with missingness dummies.
   2.1 Set missing values to the sample median. Add a dummy indicator variable.

3. **Hierarchy Principle:** Always include lower order variables when higher order variables are selected.
   3.1 If Lasso selects x^2, then always include x in the final model.
   3.2 If Lasso selects $x_1 \times x_2$, then always include x_1 and x_2 in the final model.

Best Practices Cont.

1. **Strata:** If randomization occurred at a specific level, then strata indicator variables should always be included regardless of Lasso selection.

2. **Fixed Effects:** Fixed effects, if theoretically relevant, should always be included regardless of Lasso selection.

3. When using **cross-validation** for the tuning parameter λ, always set the randomization seed.
   3.1 Cross-validation requires randomly subseting the dataset, if seeds aren't set you may get different results on different runs.

Choosing the Tuning Parameter (λ)

Analytically Derived ("Plug-in")

- Derived from asymptotic theory (Belloni et al., 2012).
- Highly conservative — i.e. heavily penalizes complexity.
- Protects against overfitting.
- **This is the default setting in Stata's `pdslasso` command.**

e.g. $\lambda = \sigma 2c\sqrt{n}\Phi^{-1}\left(1 - \frac{\gamma}{2p}\right)$

Empirical Tuning (Cross-Validation)

- Selected empirically to minimize out-of-sample MSE.
- Optimized for pure prediction tasks.
- Risk of overfitting — i.e. selecting spurious covariates.
- Generally **not recommended** for field causal analysis due to overfitting concerns.

Example Code... Still writing code... following is Preliminary

Post-Double-Selection-Example-Code

Examples I: Improved Precision

Dataset: Bertrand & Mullainathan (2004) ($n = 4{,}870$, 21 vars)

- **Situation:** RCT sends fictitious resumes with randomized Black vs. White names to employers; outcome is callback rate.
- **Problem:** Binary outcome with low base rate ($\sim$8%) means individual treatment effect estimates are noisy.
- **Why PDS:** 24 resume and employer covariates (experience, honors, industry, city, etc.) can absorb residual variance in callbacks and tighten standard errors on the race coefficient.
- **Outcome:** Lasso selected 2/21 controls, point estimates only negligibly impacted. Randomization appears totally successful.

Examples II: Differential Attrition

Dataset: RAND Health Insurance Experiment ($n = 13{,}877$, 21 vars)

- **Situation:** Families randomized into insurance plans with differing generosity (free care to 95% coinsurance); outcome is doctor visits.
- **Problem:** Completion rates ranged from 88% (free) to 63% (high coinsurance). Post-enrollment, baseline covariates reject joint balance at the 1% level.
- **Why PDS:** First Lasso on treatment selects covariates plan assignment attrition (i.e. imbalanced variables), correcting the bias from differential attrition between plans.
- **Outcome:** Lasso selected 10/21 controls, point estimates significantly impacted (0.10 $\rightarrow$ -.04) and statistical significance 1% to insignificance.

Examples III: Differential Attrition

Dataset: Project STAR ($n \approx 5{,}100$, 16 vars)

- **Situation:** Students randomized to small vs. regular K–3 classes across 79 Tennessee schools; outcome is 1st grade test scores.
- **Problem:** Over the 4-year experiment, students dropped out or switched classes at differential rates across treatment arms. After kindergarten, compliance fell as parents lobbied for class reassignment.
- **Why PDS:** Selects from 40+ demographic, teacher, and school covariates those that predict both attrition and outcomes, adjusting for attrition-induced bias in the class-size effect.
- **Outcome:** Lasso selected 6/16 controls, point estimates significantly impacted ($30.0 \to 25.3$) and statistical significance unchanged at the 1% level.

Examples IV: Observational

Dataset: Risk Factors Associated with Low Infant Birth Weight ($n = 189$, 8 vars)

- **Situation:** Estimate the effect of maternal smoking on infant birth weight.
- **Problem:** Only 189 observations — noisy treatment effect estimates.
- **How PDS helps:** Selects covariates that explain residual variance in birth weight, reducing standard errors on the smoking coefficient.
- **Outcome:** Lasso selected 2/8 controls, point estimates significantly impacted (-283.8 $\rightarrow$ -357.5) and statistical significance remained at the 1% level.

What are the Real Benefits?

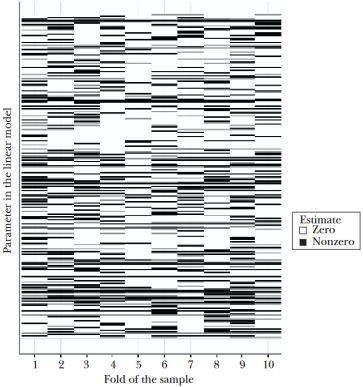
- In theory, PDS Lasso should increase model power.
  - In practice, power gains appear modest at best (Cilliers et al., 2024).
- Provides an algorithmic, transparent way to choose controls.
- Allows for the efficient processing of which controls to include when working with large dataset.

How sensitive are results to random selection?

Lasso can be sensitive to fold selection during cross-validation.

- To make results reproducible, seeds should always be set if *cross-validation* is used.
- It is **not** standard to report Lasso results over multiple folds or different seeds.
  - Demonstrating the sensitivity of your results to randomization seed or fold specification could improve transparency.

Selected Coefficients (Nonzero Estimates) across Ten Lasso Regressions



(Mullainathan & Spiess, 2017)

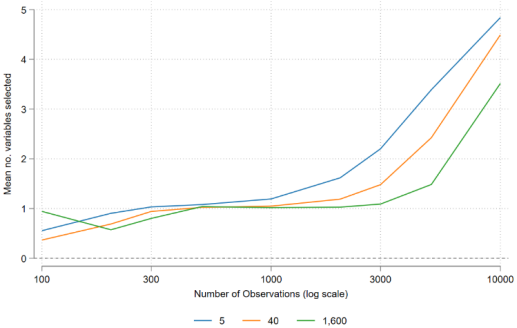
What are signs that Lasso isn't working well?

1. If convergence is taking a very long time or failing, or if results are highly unstable on model specification (i.e. fold and seed selection). This can be indicative of :
   1.1 Highly collinear covariates (Lasso is struggling to find maxima).
   1.2 Inclusion of fixed effects in interaction terms.

2. If you suspect that there are collinear variables, and think each of them has an important and distinct impact on the outcome. Then you should switch to a Ridge or ElasticNet specification.

How Many Variables to Include?

- Recommendations vary, generally it is acceptable to include all covariates at your disposal.
- Some recent work has indicated that including all covariates in Lasso selection may be suboptimal and practitioners should include only controls that have a basis in economic theory (Cilliers et al., 2024).



(a) Mean

How to Handle Missing Observations

- Impute missing values (median/mode).
- Add missingness indicator.
- Use for selection, not final regression.

```
quietly summarize x1, detail
gen x1_miss = missing(x1)
replace x1 = r(p50) if missing(x1)

pdslasso y d (x1 x1_miss)
```

How do I Handle Fixed Effects?

- Always include relevant FEs in final regression.
- Partial out FEs before Lasso selection.
- Exclude FEs from the selection step.
- Avoid FE interactions in Lasso.

```
pdslasso y d (x1 x2 x3), ///
   partial(fe lagged_y)
```

How do I Handle Multiple Treatment Variables?

- Treat each intervention as a distinct treatment variable.
- The algorithm performs the treatment-selection step for each one individually.
- Group all treatments together in the Stata syntax.
- The final model will include the union of controls selected for all treatments.

```
// Group treatments in
   parentheses
pdslasso y (t1 t2 t3) ///
   (x1 x2 x3 x4)
```

How do I Handle Heterogeneous Treatment Effects?

- Model the interaction term as a completely separate treatment.
- **Crucial:** Place the base interacting variable in the amelioration set.
- Never allow the Lasso penalty to drop the base variable.
- Do not inadvertently code the base characteristic as a treatment.

```
// t_X_z is the interaction term
// z is the base characteristic

pdslasso y (t t_X_z) ///
    (x1 x2 x3), ///
    pnotpen(z)
```

High Order Polynomial Linear Models

- Used to create a less parametric relationship.
- Stata's factor variable syntax generates polynomials and interactions.
- If using the derived λ, then the penalty parameter increases with the number of potential controls.

```
// Use c.()##c.() to create all
// main, squared, and cross terms

pdslasso y d ///
    (c.(x1 x2 x3)##c.(x1 x2 x3)), ///
    pnotpen(lagged_y strata)
```

Other Regularized Linear Models (Faul, 2020)

$\Omega(\mathbf{c})$	Regression method
$\|\mathbf{c}\|_0 = \text{number of nonzero elements in } \mathbf{c}$	*L_0 regularization*
$\|\mathbf{c}\|_1 = \sum_{m=1}^{M} \|c_m\|$	*L_1 regularization* *Least Absolute Shrinkage and Selection Operator (LASSO)*
$\frac{1}{2}\|\mathbf{c}\|^2 = \frac{1}{2}\sum_{m=1}^{M} c_m^2$	*L_2 regularization* *Ridge regression*
$\lambda\|\mathbf{c}\|_1 + \frac{1-\lambda}{2}\|\mathbf{c}\|^2$	*Elastic net regularization*

Double Machine Learning (DML) (Chernozhukov et al., 2018)

The Natural Extension of PDS Lasso

- PDS Lasso utilizes linear penalization to select covariates and orthogonalize treatment.
- **DML** extends this framework: what if the relationship is highly non-linear?
- We can use *any* ML algorithm (Random Forests, Neural Nets, etc.) to estimate the nuisance parameters.

The Partial Linear Model

$$Y_i = D_i\tau + g_0(X_i) + \epsilon_i$$

$$D_i = m_0(X_i) + V_i$$

$$Y_i = n_0(X_i) + U_i$$

DML Estimator:

$$\hat{\tau} = \left(\frac{1}{n}\sum_{i=1}^{n}\hat{V}_i\hat{V}_i\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\hat{V}_i(Y_i - \hat{n}_0(X_i))\right)$$

n_0, m_0 estimated by ML on separate folds.

Conclusion

- **An Algorithmic Defense:** PDS Lasso provides a transparent, data-driven method for selecting controls, effectively removing researcher degrees of freedom and ad-hoc specifications.
- **RCT Optimization:** It is a highly practical tool for improving precision in experimental designs, particularly when facing small sample sizes, differential attrition, or imperfect randomization.
  - Although realistically, power gains from the procedure are rarely meaningful.
- **Ease of Implementation:** Lasso is a well established technique with robust canned function in Stata, R, and Python. Any researcher can implement PDS Lasso into their workflow with minimal effort!

References I

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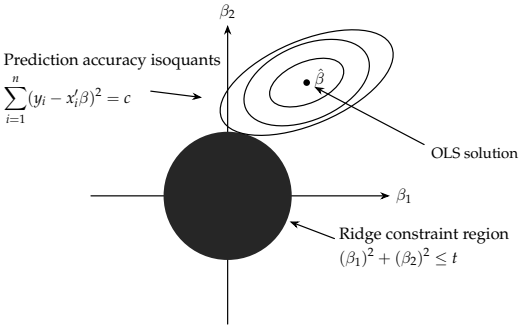
References III

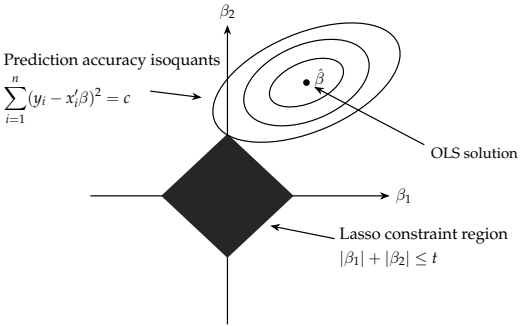
scikit-learn developers. (2025). *Cross-validation: Evaluating estimator performance* [Accessed: 2026-04-12]. https://scikit-learn.org/stable/modules/cross_validation.html

Appendix: Lasso vs Ridge Visualized

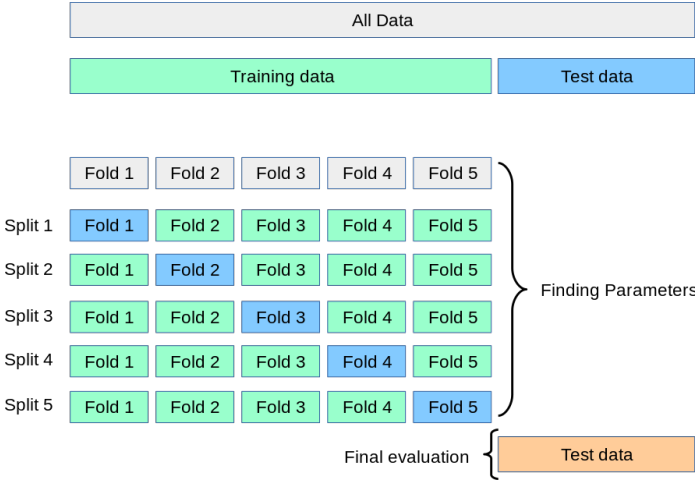
$$\hat{\beta}^{\text{ridge}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i'\beta)^2 \quad \text{s.t.} \quad (\beta_1)^2 + (\beta_2)^2 \leq t$$



$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \sum_{i=1}^{n} (y_i - x_i'\beta)^2 \quad \text{s.t.} \quad |\beta_1| + |\beta_2| \leq t$$



Cross-Validation Workflow (scikit-learn developers, 2025)



Example I: Output

	OLS	OLS+Controls	PDS-Lasso
African-American Name	-0.0320***	-0.0312***	-0.0317***
	(0.0078)	(0.0077)	(0.0077)

Standard errors in parentheses

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Example II: Output

	OLS	OLS+Controls	PDS-Lasso
Zero Copay Plan	0.1038***	-0.0302	-0.0365
	(0.0204)	(0.0337)	(0.0335)

Standard errors in parentheses

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Example III: Output

	OLS	OLS+Controls	PDS-Lasso
Small Class Assignment	29.9701***	23.8968***	25.3256***
	(4.8006)	(4.6211)	(4.3507)

Standard errors in parentheses

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Example IV: Output

	OLS	OLS+Controls	PDS-Lasso
smoke	-283.7767***	-352.0445***	-357.5413***
	(103.9132)	(105.7858)	(107.0745)

Standard errors in parentheses

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$